Enhanced EEG Imagined Digit Recognition System — Experimental Summary

Evaluation Metric / Attribute	Enhanced Approach Value	Benchmark / Context
Participants Evaluated	**23 Subjects (Subject 0 to 24)**	All available subjects with full 14 channels
Total EEG Recordings (Trials)	**230 EDF recordings**	10 digit trials (0-9) per subject
Total Evaluated Windows	**120,265 windows**	50 ms duration with 50% overlap (25 ms hop)
Feature Vector Dimensionality	**182-D features per window**	56 Time-Domain + 70 Band-Power + 42 Hjorth +
Window-Level Accuracy (Mean ± SD)	**83.28% ± 12.72%**	Across 23 participants (10-fold positional CV)
Window-Level Pooled Accuracy	**83.34%**	+14.84% gain over Kumar et al. (68.50%)
Window-Level Macro-F1	**83.34%**	Balanced across all 10 digit classes (0-9)
Corrected Trial-Level Accuracy (Mean ± SD)	**99.83% ± 0.82%**	Majority voting over constituent trial windows
Corrected Trial-Level Pooled Accuracy	**99.83%**	229/230 trials correctly classified
Corrected Trial-Level Macro-F1	**99.83%**	Near-perfect trial-level discrimination
Cross-Validation Strategy	**Within-subject 10-fold positional CV**	Zero temporal leakage (stride-modulo split)